Zhaoxiang (Aaron) Feng

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Education

University of California San Diego (UCSD) *Sep. 2022 – Apr. 2026*
B.S. in Data Science & B.S. in Probability & Statistics *(expected)*

- GPA: 3.83 (Major GPAs: 3.95 in Data Science, 3.93 in Statistics)
- Honors: Provost Honors (multiple quarters)
- Graduate-level Coursework: Advanced Time Series Analysis, Machine Learning, Applied Statistics, Privacy-sensitive Data Systems

Research Interests

I am interested in building AI agents with persistent **memory** that enables test-time continual learning: systems that accumulate and reuse concepts, self-verify, and improve after deployment. My focus areas include **memory** architectures, calibrated uncertainty, and efficient inference.

Publications

[1] Matthew Ho, Chen Si, **Zhaoxiang Feng**, Fangxu Yu, Yichi Yang, Zhijian Liu, Zhiting Hu, Lianhui Qin. "ArcMemo: Abstract Reasoning Composition with Lifelong LLM Memory."
***Runner Up**, ARC Prize 2025 Paper Awards — Top 8 of 90 paper submissions.* *[paper] [code]*

[2] **Zhaoxiang Feng**, David Scott Lewis, Enrique Zueco. "LabMemo: Concept-Level Memory for Autonomous Scientific Discovery."
Accepted at IEEE IROS 2025 Workshop on Embodied AI and Robotics for Future Scientific Discovery (AIR4S) (non-archival), 2025. *[paper] [code]*

[3] **Zhaoxiang Feng**, David Scott Lewis. "SOKRATES: Distilling Symbolic Knowledge into Option-Level Reasoning via Solver-Guided Preference Optimization."
Accepted at AAAI 2026 Bridge Program on Logic & AI: Logical and Symbolic Reasoning in Language Models (LMReasoning), 2026. *[paper] [code]*

[4] **Zhaoxiang Feng**, Mingyang Yao, Charlie Sun, David Scott Lewis. "Which Memory Operation Drives Recovery? A Factorial Study of Retrieve, Write, and Manage Adaptation under Domain Shift."
Accepted at ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving (LLA), 2026. *[paper]*

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[5] Jinzhou Tang, Yufan Zhou, Zixuan Wang, **Zhaoxiang Feng**, Xinle Yu, Steven Ngo, Zhengding Hu, Luoshang Pan, Lianhui Qin, Yufei Ding, Tianmin Shu, Jingbo Shang, Zhiting Hu, Zhen Wang. "S^3-Sim: Simulating Humans for Personalized Language Modeling."
Under review at International Conference on Machine Learning (ICML), 2026.

[6] **Zhaoxiang Feng**, Zhongyan Luo, Mingyang Yao, Charlie Sun. "Explanation-Consistency Graphs: Neighborhood Surprise in Explanation Space for Training Data Debugging."
Under review at ACL Rolling Review (ARR), 2026.

[7] **Zhaoxiang Feng**, Zhongyan Luo, Mingyang Yao, Charlie Sun, Yuwen Zhang, David Scott Lewis. "EP-Prior: Interpretable ECG Representations via Electrophysiology Constraints."
Under review at IJCAI-ECAI 2026 AI and Health Track, 2026.

[8] **Zhaoxiang Feng**, Mingyang Yao, Shuheng Cao, Young Su Ko, Wei Wang. "TrustPPI: Deformation Stability as a Model-Agnostic Trust Signal for Protein-Protein Interaction Prediction."
Under review at KDD 2026 AI4Sciences Track, 2026. *[code]*

Research Experience

Research Assistant | Q-Lab, UC San Diego *Jan. 2025 – Present*
Advisors: Prof. Lianhui Qin and Matthew Ho

- Led design of ArcMemo's program-synthesis-style memory ontology: many-to-one puzzle-to-feature mappings and curated concept parameterizations for concept-level reasoning.
- Engineered a reasoning-based retrieval mechanism (System-2 exploration) to resolve embedding failures, achieving a 7.5% relative gain on ARC-AGI-1 (59.33% official score).
- Built a concept dataset generation pipeline transforming hand-written concepts into validated helper puzzles via multi-stage LLM generation, code synthesis, and automated testing.
- Extended to AIME math problems via a metacognitive self-assessment pipeline, improving accuracy by 9.3% through self-reflective memory usage.

Research Assistant | Wang Lab, UC San Diego *Jun. 2025 – Present*
Advisors: Prof. Wei Wang and Young Su Ko

- Developing TrustPPI: a model-agnostic trust signal for PPI prediction via deformation stability — injecting Gaussian noise (σ=0.10, K=10) into intermediate embeddings and combining flip-rate and variance signals; validated across 4 architectures (CNN, Transformer, ESM-2, E(3)-GNN) and 11 model×species combinations.
- Achieved 0.66–0.83 AUROC for error detection across all models, with trust-guided selection discovering true interactions 2–3.5× faster than confidence baselines and +2–10% accuracy gain at 80% coverage.
- Architected a heterogeneous Mixture-of-Experts system for chemical reaction prediction on USPTO (1M+ reactions) using directed message-passing GNN encoders with shortest-path positional encodings and learned routing mechanisms across four specialized expert models.

Student Researcher | Halıcıoğlu Data Science Institute, UC San Diego *Sep. 2025 – Present*
Advisor: Prof. Hao Zhang (Senior Capstone)

- Built an end-to-end distributed training stack for a 1.8B-parameter language model on 8 NVIDIA B200 GPUs, implementing pipeline parallelism and scaling analysis.
- Ported DFlash (diffusion-based speculative decoding) to JAX/TPU, achieving 3.02× average speedup (4.93× on math, 9 benchmarks); exploited TPU MXU 128×128 tile structure for flat verification cost up to K=128, lifting the GPU-motivated K=16 constraint; integrated into vLLM with 2.31× end-to-end speedup; PR-ready for vllm-project/tpu-inference.

Student Researcher | Halıcıoğlu Data Science Institute, UC San Diego *Sep. 2025 – Present*
Advisor: Prof. Yian Ma (Senior Capstone)

- Designed MI-based uncertainty signals for epistemic uncertainty detection and risk-controlled stopping in LLM reasoning.
- Developing an uncertainty-guided interactive reasoning framework using self-revision mutual information for ask-vs-answer stopping; designed Evidence Sensitivity and Hypothesis Discrimination probes with Clopper-Pearson risk-controlled calibration, improving AR-Bench Detective Cases from 36% to 45%.

Research Assistant | U.S. Immigration Policy Center, UC San Diego *Sep. 2023 – Jun. 2024*
Advisors: Prof. Tom K. Wong and Dr. Gabriel De Roche

- Built ARIMA and exponential smoothing forecasting models on 20+ years of naturalization data for 2024 election policy briefs; developed automated data pipelines and Tableau dashboards for non-technical stakeholders.

Teaching & Service

Tutor & Grader *Fall 2024 – Present*
University of California San Diego

- **Tutor:** DSC 40A: Theoretical Foundations of Data Science I (Summer 2025)
- **Grader:** MATH 180A/C, MATH 181A, MATH 185 (Fall 2024 – Fall 2025, 7 quarters)

Technical Skills

Languages: Python, R, SQL, Java, JavaScript, HTML

ML & LLM: PyTorch, PyTorch Geometric, Transformers, scikit-learn, LLM fine-tuning and post-training, RAG systems, memory-augmented LLMs, reasoning-based retrieval, continual learning

Data & Infrastructure: Pandas, Spark, AWS, Docker, distributed training, vector databases